

Total No. of Questions : 8]

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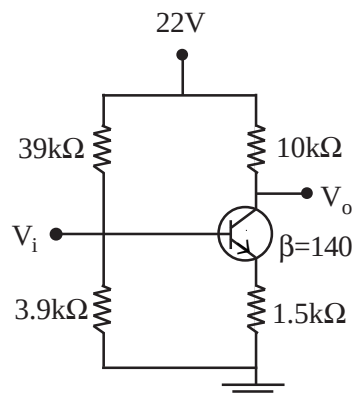
Roll No

CS/IT-304 (GS)
B.E. III Semester Examination, June 2020
Grading System (GS)
Electronics Devices and Circuits
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Draw the circuit diagram and explain the principle of operation of the following oscillators
 - i) RC phase shift oscillator
 - ii) LC oscillator
2.
 - a) Explain how a Zener diode can be used as a voltage regulator.
 - b) Write a short note on UPS.
3. Explain the operation of positive clipper with suitable waveforms.
4.
 - a) Explain the working of Monostable multivibrators.
 - b) What do you understand by SMPS?
5. Determine the DC bias voltage V_{CE} and I_C for the voltage divider configuration of fig.



6. Draw and explain voltage to current and current to voltage convertor.

OR

Describe Schmitt trigger.

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7. Draw and explain circuit of a transformer coupled single stage transistor power amplifier and explain the need for impedance matching.

OR

“A depletion mode MOSFET can also be operated in enhancement mode but an enhancement mode MOSFET cannot be operated in depletion mode.” Justify

8. Answer any four of the following.
- a) Write a note on Zener Diode.
 - b) What are the feedback techniques applied in a feedback amplifier? Explain.
 - c) Write a note on clamper.
 - d) Differentiate between an integrator and differentiator.
 - e) Explain in brief the working of a series regulated power supply.
 - f) What is meant by Minimum Load Resistance of a power supply.
